

Beautiful Permutation II

Time limit: 1.00 s

Memory limit: 512 MB

A permutation of integers $1, 2, \dots, n$ is called *beautiful* if there are no adjacent elements whose difference is 1.

Given n , construct the lexicographically minimal beautiful permutation if such a permutation exists.

Input

The only line contains an integer n .

Output

Print the lexicographically minimal beautiful permutation of integers $1, 2, \dots, n$. If there is no such permutation, print "NO SOLUTION".

Constraints

- $1 \leq n \leq 10^6$

Example 1

Input	Output
5	1 3 5 2 4

Example 2

Input	Output
3	NO SOLUTION